Communications in Theoretical Physics

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To cite this article: Shao Cheng-Gang *et al* 2006 *Commun. Theor. Phys.* **45** 499

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Commun. Theor. Phys. (Beijing, China) **45** (2006) pp. 499–504
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Vol. 45, No. 3, March 15, 2006

Casimir Effect for Dielectric Plates*

SHAO Cheng-Gang, TONG Ai-Hong, and LUO Jun†

Department of Physics, Huazhong University of Science and Technology, Wuhan 430074, China

(Received May 30, 2005; Revised July 12, 2005)

Abstract *We generalize Kupisewska method to the three-dimensional system and another derivation of the Casimir effect between two dielectric plates is presented based on the explicit quantization of the electromagnetic field in the presence of dielectrics, where the physical meaning of "evanescent mode" is discussed. The Lifshitz's formula is rederived using all the vacuum mode functions, which include the contribution of the 'evanescent modes'. Only in the case of the perfect metallic plates will the evanescent modes become unimportant.*

PACS numbers: 12.20.Ds
Key words: Casimir effect, electromagnetic field quantization, evanescent mode

1 Introduction

During the last few years the Casimir effect has attracted a lot of experimental and theoretical attention as a nontrivial macroscopic evidence for the existence of zero-point oscillations of electromagnetic field.[1−4] This is in part due to the resurgence of the interest in the fundamental problems connected with the physical vacuum. Except for the familiar mode summation method of deriving Casimir forces as employed by Casimir himself,[5] various concepts and calculational techniques have been developed, but only a few of them have turned out to be capable of dealing with realistic dielectric bodies.

Lifshitz first introduced a macroscopic theory for the evaluation of the Casimir force between two dielectric plates by calculating the Maxwell stress tensor, where the electromagnetic field and the material bodies are treated macroscopically, and explicit field quantization is avoided.[6] The dielectric media are characterized by randomly fluctuating sources of current densities having a δ-function correlation. Lifshitz's theory is very complicated and developed by many authors, such as Schwinger,[7,8] Matloob,[9−11] and Tomas.[12] In these methods, the dissipation-fluctuation relation or the field correlation function plays an important role. Another derivation of the Lifshitz results based on the surface mode is also discussed by some authors, starting directly from the zero-point energy of electromagnetic field.[1,13]

Kupisewska has presented an instructive treatment of the Casimir effect between dielectric plates,[14,15] where the electromagnetic field is quantized in the presence of dielectric material and the notion of physical photons is introduced. The Casimir force can be calculated from the pressure of the electromagnetic field in the vacuum state on the slab with the help of the Maxwell stress tensor. In this approach, the difficulty is in quantizing the electromagnetic field in the presence of dielectric material, where the physical meaning of "evanescent mode" is not clear. The intricate structure of the field operators has restricted approach to one-dimensional systems so far.[11,12,16−18]

In the present paper, we formally generalize Kupisewska method to the three-dimensional system. A rigorous derivation of Casimir effect using three-dimensional mode functions, including the contribution of the evanescent modes, is given and the results are the same as the Lifshitz's formula . We also discuss the physical meaning of evanescent mode and find that only in the case of the perfect metallic plates, the evanescent modes will become unimportant.

2 Quantization of Electromagnetic Field and Physical Meaning of "Evanescent Mode"

The most popular interpretation of the Casimir force is based upon the electromagnetic vacuum pressure.[19,20] The magnetic intensity $\vec{B}$ and the electric intensity $\vec{E}$ can be expressed by the vector potential $\vec{A}(\vec{r},t)$ in the relations $\vec{B}=\nabla\times\vec{A}$, $\vec{E}=-\partial\vec{A}/\partial t$ under the Coulomb gauge $\nabla\cdot\varepsilon\vec{A}=0$, where $\varepsilon(\vec{r})$ is the dielectric function. The equation of motion for the vector potential is

$$\nabla\times\nabla\times\vec{A}^{+}(\vec{r},\omega)-\varepsilon(\vec{r},\omega)\frac{\omega^2}{c^2}\vec{A}^{+}(\vec{r},\omega)=0 \qquad (1)$$

with $\vec{A}(\vec{r},t)=\int_0^\infty(\mathrm{d}\omega/2\pi)\vec{A}^{+}(\vec{r},\omega)\,\mathrm{e}^{-\mathrm{i}\omega t}+\mathrm{c.c.}$

Let us consider the general process of the electromagnetic field quantization. Roughly speaking, there have been two routes to study the behavior of the quantized electromagnetic field in the presence of dielectrics. One is to use the Green-tensor formation by employing the dissipation-fluctuation theory,[21,22] which has successfully been applied to the study of the Casimir effect.[12] The other is to perform explicit field quantization. Here we pay attention on the second one. By introducing the annihilation and the creation operators $b_{k\lambda}$ and $b_{k\lambda}^{\dagger}$, the vector potential operator can be expanded as a linear combination of the quantum vacuum mode functions,

$$\vec{A}(\vec{r},t)=\sum_{k,\lambda}c\Big(\frac{\hbar}{2\omega_k\varepsilon_0}\Big)^{1/2}\big[b_{k\lambda}\,\mathrm{e}^{-\mathrm{i}\omega t}\vec{f}_{k\lambda}(\vec{r}) + b_k^{\dagger}\,\mathrm{e}^{\mathrm{i}\omega t}\vec{f}_{k\lambda}^{*}(\vec{r})\big]\,. \qquad (2)$$

*The project partially supported by the State Key Basic Research Program of China under Grant No. 2003CB716300 and National Natural Science Foundation of China under Grant No. 10121503

†E-mail: junluo@mail.hust.edu.cn

The operators $b_{k\lambda}$ and $b^{\dagger}_{k\lambda}$ act in the Fock space. The physical photons are associated with the mode functions $\vec{f}_{k\lambda}(\vec{r})$. The photon frequency is ω_k and polarization labels by $\lambda = s$ (s-polarized or TE) and by $\lambda = p$ (p-polarized or TM). The mode functions obey the equation

$$\nabla \times \nabla \times \vec{f}_{k\lambda}(\vec{r}) - \frac{\omega_k^2 \varepsilon(\vec{r})}{c^2} \vec{f}_{k\lambda}(\vec{r}) = 0\,, \tag{3}$$

together with gauge condition and orthonormality conditions:

$$\nabla \cdot [\varepsilon(\vec{r}) \vec{f}_{k\lambda}(\vec{r})] = 0\,, \tag{4}$$

$$\int \mathrm{d}^3\vec{r} [\varepsilon(\vec{r}) \vec{f}_{k\lambda}(\vec{r}) \vec{f}_{-k'\lambda'}(\vec{r})] = \delta_{kk'}\delta_{\lambda\lambda'}\,. \tag{5}$$

In free space, $\vec{f}_{k\lambda} = (1/\sqrt{V})\vec{e}_{\lambda}\,\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{r}}$, V is the normalization volume, $\vec{e}_{\lambda}$ is the unit polarization vector, which is perpendicular to the wave vector $\vec{k}$, apparently.

Let us apply the above considerations to the configuration with two parallel infinite dielectric plates each of thickness d, a distance of a apart (Fig. 1). The two dielectric plates situated in (y, z) planes has the frequency-dependent dielectric permittivity ε. The complex refractive index is assumed to be equal to n with $n^2 = \varepsilon$. Then the vector potential may be written as

$$\vec{A}(\vec{r}, t) = \sum_{\lambda} c \frac{1}{(2\pi)^3} \int \mathrm{d}^3\vec{k} \Big(\frac{\hbar}{2\omega_k \varepsilon_0}\Big)^{1/2} \times \vec{e}_{\lambda} \Big[b_{k\lambda}\,\mathrm{e}^{-\mathrm{i}\omega t} \vec{f}_{k\lambda}(\vec{r}) + b^{\dagger}_{k\lambda}\,\mathrm{e}^{\mathrm{i}\omega t} \vec{f}_{k\lambda}{}^{*}(\vec{r}) \Big]\,. \tag{6}$$

The parameter $\lambda = s$, and p labels the two possible polarizations. If ε is constant in the region of the dielectric, then the mode function satisfies the equation

$$\nabla^2 \vec{f}_{k\lambda}(\vec{r}) + \frac{\omega_k^2 \varepsilon(\vec{r})}{c^2} \vec{f}_{k\lambda}(\vec{r}) = 0 \tag{7}$$

with the appropriate boundary condition and the orthonormality conditions.

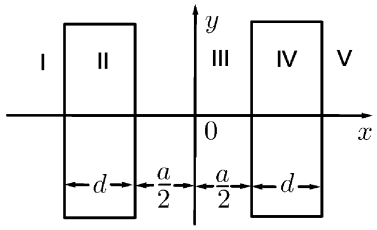


Fig. 1 Configuration of the dielectric plates with permittivity $\varepsilon(\vec{r})$.

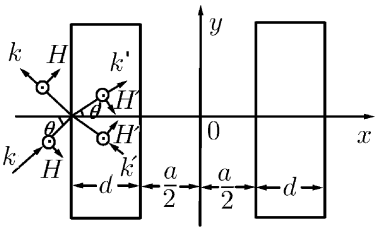


Fig. 2 The TE modes polarization $\lambda = s$. Electric vector $\vec{E}$ points along z axis.

Exploiting the translational and rotational symmetry in the y–z plane, we may select the incident wave vector in the x–y plane for simplicity, and the angle between the incident direction and the x axis is θ. Therefore, we may write the mode function in the form of $\vec{f}_{k\lambda}(\vec{r}) = f_{k\lambda}(x)\,\mathrm{e}^{\mathrm{i}k_y y}$ with $k_y = k\sin\theta$. $f_{k\lambda}(x)$ is given in Table 1. θ' is confirmed by the refraction law $\sin\theta = n\sin\theta'$.

Table 1 Mode functions $f_{k\lambda}(x)$.

$k\cos\theta = k_x > 0$	$k\cos\theta = k_x < 0$	
$\mathrm{e}^{\mathrm{i}k\cos\theta x} + R_{k\lambda}\,\mathrm{e}^{-\mathrm{i}k\cos\theta x}$	$T_{-k\lambda}\,\mathrm{e}^{\mathrm{i}k\cos\theta x}$	$x \in \left(-\infty, -\frac{a}{2} - d\right]$
$A_{k\lambda}\,\mathrm{e}^{\mathrm{i}kn\cos\theta' x} + B_{k\lambda}\,\mathrm{e}^{-\mathrm{i}kn\cos\theta' x}$	$E_{-k\lambda}\,\mathrm{e}^{\mathrm{i}kn\cos\theta' x} + F_{-k\lambda}\,\mathrm{e}^{-\mathrm{i}kn\cos\theta' x}$	$x \in \left[-\frac{a}{2} - d, -\frac{a}{2}\right]$
$C_{k\lambda}\,\mathrm{e}^{\mathrm{i}k\cos\theta x} + D_{k\lambda}\,\mathrm{e}^{-\mathrm{i}k\cos\theta x}$	$C_{-k\lambda}\,\mathrm{e}^{\mathrm{i}k\cos\theta x} + D_{-k\lambda}\,\mathrm{e}^{-\mathrm{i}k\cos\theta x}$	$x \in \left[-\frac{a}{2}, \frac{a}{2}\right]$
$E_{k\lambda}\,\mathrm{e}^{\mathrm{i}kn\cos\theta' x} + F_{k\lambda}\,\mathrm{e}^{-\mathrm{i}kn\cos\theta' x}$	$A_{-k\lambda}\,\mathrm{e}^{\mathrm{i}kn\cos\theta' x} + B_{-k\lambda}\,\mathrm{e}^{-\mathrm{i}kn\cos\theta' x}$	$x \in \left[\frac{a}{2}, \frac{a}{2} + d\right]$
$T_{k\lambda}\,\mathrm{e}^{\mathrm{i}k\cos\theta x}$	$\mathrm{e}^{\mathrm{i}k\cos\theta x} + R_{-k\lambda}\,\mathrm{e}^{-\mathrm{i}k\cos\theta x}$	$x \in \left[\frac{a}{2} + d, \infty\right)$

When the electric field intensity is perpendicular to the incident plane as shown in Fig. 2, polarization direction is in z axis, labeling by $\lambda = s$. The boundary conditions of E_z and H_y at the points $x = -a/2 - d$, $-a/2$, $a/2$, and $a/2 + d$ correspond to a system of eight linear equations,

$$\mathrm{e}^{-\mathrm{i}k(\cos\theta)(a/2+d)} + R_{ks}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a/2+d)} = A_{ks}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')(a/2+d)} + B_{ks}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')(a/2+d)}\,,$$

$$\cos\theta(\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)(a/2+d)} - R_{ks}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a/2+d)}) = n\cos\theta'(A_{ks}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')(a/2+d)} - B_{ks}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')(a/2+d)})\,,$$

$$A_{ks}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')a/2} + B_{ks}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')a/2} = C_{ks}\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)a/2} + D_{ks}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)a/2}\,,$$

$$n\cos\theta'(A_{ks}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')a/2} - B_{ks}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')a/2}) = \cos\theta(C_{ks}\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)a/2} - D_{ks}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)a/2})\,,$$

$$
\begin{aligned}
&C_{ks}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)a/2}+D_{ks}\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)a/2}=E_{ks}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')a/2}+F_{ks}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')a/2}\,,\\
&\cos\theta(C_{ks}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)a/2}-D_{ks}\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)a/2})=n\cos\theta'(E_{ks}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')a/2}-F_{ks}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')a/2})\,,\\
&E_{ks}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')(a/2+d)}+F_{ks}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')(a/2+d)}=T_{ks}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a/2+d)}\,,\\
&n\cos\theta'(E_{ks}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')(a/2+d)}-F_{ks}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')(a/2+d)})=\cos\theta T_{ks}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a/2+d)}\,.
\end{aligned}
\tag{8}
$$

The coefficients R_{ks}, T_{ks}, A_{ks}, B_{ks}, C_{ks}, D_{ks}, E_{ks}, and F_{ks} are

$$
\begin{aligned}
R_{ks}&=\frac{r_s+r_s\,\mathrm{e}^{2ik(\cos\theta)a}(\,\mathrm{e}^{2\mathrm{i}kn(\cos\theta')d}-r_{ds}^2)/(1-r_{ds}^2\,\mathrm{e}^{2\mathrm{i}kn(\cos\theta')d})}{1-r_s^2\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a}}\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)(a+2d)}\,,\\
A_{ks}&=\frac{t_s(1+r_{ds}r_s\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a})}{t_{ds}(1-r_s^2\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a})}\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)(a/2+d)}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')a/2}\,,\\
B_{ks}&=\frac{t_s(r_{ds}+r_s\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a})}{t_{ds}(1-r_s^2\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a})}\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)(a/2+d)}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')a/2}\,,\\
C_{ks}&=\frac{t_s\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)d}}{1-r_s^2\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a}}\,,\qquad D_{ks}=\frac{t_sr_s\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a-d)}}{1-r_s^2\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a}}\,,\\
E_{ks}&=\frac{t_s^2}{t_{ds}(1-r_s^2\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a})}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a/2+d)}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')(a/2+d)}\,,\\
F_{ks}&=\frac{t_s^2r_{ds}}{t_{ds}(1-r_s^2\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a})}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a/2+d)}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')(a/2+d)}\,,\\
T_{ks}&=\frac{t_s^2\,\mathrm{e}^{-2\mathrm{i}k(\cos\theta)d}}{1-r_s^2\,\mathrm{e}^{2\mathrm{i}k(\cos\theta)a}}\,,
\end{aligned}
\tag{9}
$$

with r_s and t_s the reflection and refraction coefficients for one dielectric plate

$$
\begin{aligned}
r_s&=\frac{r_{ds}(\,\mathrm{e}^{2\mathrm{i}kn(\cos\theta')d}-1)}{1-r_{ds}^2\,\mathrm{e}^{2\mathrm{i}kn(\cos\theta')d}}\,,\quad t_s=\frac{4n\cos\theta'\cos\theta}{(n\cos\theta'+\cos\theta)^2}\,\frac{\mathrm{e}^{\mathrm{i}kn(\cos\theta')d}}{1-r_{ds}^2\,\mathrm{e}^{2\mathrm{i}kn(\cos\theta')d}}\,,\\
r_{ds}&=\frac{n\cos\theta'-\cos\theta}{n\cos\theta'+\cos\theta}\,,\quad t_{ds}=\frac{2n\cos\theta'}{n\cos\theta'+\cos\theta}\,.
\end{aligned}
\tag{10}
$$

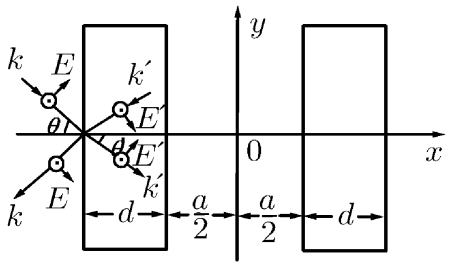


Fig. 3 The TM modes polarization $\lambda = p$. Magnetic vector $\vec{H}$ points along z axis.

When the electric field intensity is parallel to the incident plane as shown in Fig. 3, the magnetic field intensity points along z axis and the polarization labels by $\lambda = p$. The boundary conditions of E_y and H_z at the points $x = -a/2 - d$, $-a/2$, $a/2$, and $a/2 + d$ also correspond to a system of eight linear equations. Here we only write down two equations corresponding to the plane of $x = -a/2 - d$,

$$
\begin{aligned}
&e^{-\mathrm{i}k(\cos\theta)(a/2+d)}+R_{kp}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a/2+d)}=n(A_{kp}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')(a/2+d)}+B_{kp}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')(a/2+d)})\,,\\
&\cos\theta(\,\mathrm{e}^{-\mathrm{i}k(\cos\theta)(a/2+d)}-R_{kp}\,\mathrm{e}^{\mathrm{i}k(\cos\theta)(a/2+d)})=\cos\theta'(A_{kp}\,\mathrm{e}^{-\mathrm{i}kn(\cos\theta')(a/2+d)}-B_{kp}\,\mathrm{e}^{\mathrm{i}kn(\cos\theta')(a/2+d)})\,.
\end{aligned}
\tag{11}
$$

The coefficients R_{kp}, T_{kp}, A_{kp}, B_{kp}, C_{kp}, D_{kp}, E_{kp}, and F_{kp} have the same form as Eq. (9), by replacing the subscript label $\lambda = s$ by $\lambda = p$ with

$$
\begin{aligned}
r_p&=\frac{r_{dp}(\,\mathrm{e}^{2ikn(\cos\theta')d}-1)}{1-r_{dp}^2\,\mathrm{e}^{2\mathrm{i}kn(\cos\theta')d}}\,,\quad t_p=\frac{4n\cos\theta'\cos\theta}{(\cos\theta'+n\cos\theta)^2}\,\frac{\mathrm{e}^{\mathrm{i}kn(\cos\theta')d}}{1-r_{dp}^2\,\mathrm{e}^{2\mathrm{i}kn(\cos\theta')d}}\,,\\
r_{dp}&=\frac{\cos\theta'-n\cos\theta}{\cos\theta'+n\cos\theta}\,,\quad t_{dp}=\frac{2n\cos\theta'}{\cos\theta'+n\cos\theta}\,.
\end{aligned}
\tag{12}
$$

The results derived above have only considered the normal modes of $k_x^2 = k^2p^2 > 0$ with the parameter $p = \cos\theta$. Moreover, additional kinds of modes of $k_x^2 = k^2p^2 < 0$ also exist, which may be called 'evanescent modes'. For these modes the inequality $k^2 < k_y^2 + k_z^2 = k_{||}^2$ holds and p can take the imaginary number. The term 'evanescent' means that the reflection and transmission waves are exponentially damping for $x < -a/2 - d$ and $x > a/2 + d$. However, according to Table 1, the incident wave $\mathrm{e}^{-\kappa x}$ is divergent for $x \to -\infty$ if $k_x = \mathrm{i}\kappa$, $\kappa > 0$. This divergence can be avoided by using the formalism of surface modes,[1] where the mode function behaves like $\mathrm{e}^{-\kappa|x|}$ and is exponentially damping for $|x| \to \infty$, κ may take discrete values. These surface modes in physics describe wave propagating parallel to the surface of the plates. However, any exponentially damping (or exponentially increasing) mode functions cannot be normality in the infinite volume $V \to \infty$, which is not consistent with the orthonormality condition (5). In physical sense, it is well known that the k_x space and the x space are physically equivalent in quantum theory. If momentum k_x takes the imaginary value $k_x = \mathrm{i}\kappa, \kappa > 0$, position x should also be replaced by $\chi = \mathrm{i}x$. The evanescent mode function outside the plates can be written as

$$f_k(x) = f_\kappa(\chi) = \begin{cases} \mathrm{e}^{\mathrm{i}\kappa\chi} + R\,\mathrm{e}^{-\mathrm{i}\kappa\chi}, & \chi < -\dfrac{a}{a} - d \\ T\,\mathrm{e}^{\mathrm{i}\kappa\chi}, & \chi > \dfrac{a}{a} + d \end{cases} \quad \text{with} \quad k_x = \mathrm{i}\kappa, \quad x = -\mathrm{i}\chi\,. \tag{13}$$

In the above 'evanescent modes' is not evanescent in the space χ. The integral of the orthonormality conditions (5) can be taken in the space χ. In mathematical sense, the evanescent modes and the normal modes can be expressed in the same form as in Table 1, which form a complete set of functions in the space satisfying the orthonormality conditions.

The formal definition (13) is a simple generalization of the normal mode function, which did not affect the distribution,

$$\sum_\lambda \frac{1}{(2\pi)^3}\int \mathrm{d}k_y \int \mathrm{d}k_z \int \mathrm{d}k_x \vec{f}_{k\lambda}(\vec{r})\vec{f}_{k\lambda}{}^*(\vec{r}\,') = \sum_\lambda \vec{e}_\lambda\vec{e}_\lambda\delta^3(\vec{r} - \vec{r}\,')\,. \tag{14}$$

Using the commutation relations

$$[b_{k\lambda}, b^\dagger_{k'\lambda'}] = (2\pi)^3\delta_{\lambda\lambda'}\delta(k_x - k_{x'})\delta(k_y - k_{y'})\delta(k_z - k_{z'})\,, \tag{15}$$

we get equal-time commutation relations between the vector potential operator and its conjugate momentum,

$$[\vec{A}(\vec{r}, t), -\varepsilon_0\vec{E}(\vec{r}, t)] = \mathrm{i}\hbar\sum_\lambda \vec{e}_\lambda\vec{e}_\lambda\delta^3(\vec{r} - \vec{r}\,')\,. \tag{16}$$

3 Casimr Force for Dielectric Plates

With the explicit form of the mode functions given, we may derive the interaction between two dielectric plates as the effect of vacuum pressure, which may be expressed by Maxwell tensor,[14,19]

$$T^{ij} = -\frac{1}{2}\Big[2\varepsilon(\vec{r})\varepsilon_0 E^i E^j + 2\frac{1}{\mu_0}B^iB^j - (\vec{E}^{\,2} + \vec{B}^{\,2})\delta_{ij}\Big]\,. \tag{17}$$

The vacuum field correlations $\langle E^iE^j\rangle$ and $\langle B^iB^j\rangle$ are built by taking the appropriate derivatives of the vector potential operator. The vacuum pressure on a body can be calculated by integrating of T^{ij} over the surrounding surface of the volume,

$$f^i = \int_\Sigma T^{ij}n^j\,\mathrm{d}s\,, \tag{18}$$

where n^j is normal to the integral surface. In our case of two parallel plates the pressure will be the difference between T_{xx} component of the stress tensor outside the plates and that between the plates, calculated on the surface:

$$f = T_{xx}\Big(x = -\frac{a}{2} - d\Big) - T_{xx}\Big(x = -\frac{a}{2}\Big)\,. \tag{19}$$

Here, $T_{xx} = [\varepsilon_0(E_y^2 + E_z^2 - E_x^2) + (1/\mu_0)(B_y^2 + B_z^2 - B_x^2)]/2$. For the single mode function with the angel θ between the wave vector and the dielectric surface, we have $T_{xx} = [\varepsilon_0E^2 + (1/\mu_0)B^2(1 - 2\sin^2\theta)]$ for $\lambda = s$ and $T_{xx} = [\varepsilon_0E^2(1 - 2\sin^2\theta) + (1/\mu_0)B^2]/2$ for $\lambda = p$. The electromagnetic field operators expand as a linear combination of mode functions. As an example let us calculate the T_{xx} in region I, which is equal to that in region V.

$$\vec{E}(\vec{r}, t) = \sum_\lambda \frac{1}{(2\pi)^3}\int_{-\infty}^{\infty} \mathrm{d}^2\vec{k}_\perp\Big[\int_{\mathrm{i}k_\perp\to 0\to\infty} \mathrm{d}k_x\Big(\frac{\hbar}{2\omega_k\varepsilon_0}\Big)^{1/2}\vec{e}_\lambda b_{k\lambda}\mathrm{i}\omega_k\,\mathrm{e}^{-\mathrm{i}\omega_k t}\,\mathrm{e}^{\mathrm{i}\vec{k}_\perp\cdot\vec{r}}(\,\mathrm{e}^{\mathrm{i}k_x x} + R_{k\lambda}\,\mathrm{e}^{-\mathrm{i}k_x x}) + \mathrm{c.c.}\Big]$$

$$+\sum_\lambda \frac{1}{(2\pi)^3}\int_{-\infty}^{\infty} d^2\vec{k}_\perp\Big[\int_{-\infty\to 0\to -\mathrm{i}k_\perp} \mathrm{d}k_x\Big(\frac{\hbar}{2\omega_k\varepsilon_0}\Big)^{1/2}\vec{e}_\lambda b_{k\lambda}\mathrm{i}\omega_k\,\mathrm{e}^{-\mathrm{i}\omega_k t}\,\mathrm{e}^{\mathrm{i}\vec{k}_\perp\cdot\vec{r}}T_{-k\lambda}\,\mathrm{e}^{\mathrm{i}k_x x} + \mathrm{c.c.}\Big]\,,$$

$$\vec{B}(\vec{r},t)=\frac{1}{(2\pi)^3}\int_{-\infty}^{\infty}\mathrm{d}^2\vec{k}_\perp\Big[\int\limits_{ik_\perp\to 0\to\infty}\mathrm{d}k_x\Big(\frac{\hbar}{2\omega_k\varepsilon_0}\Big)^{1/2}\vec{e}_p b_{ks}\mathrm{i}k\,\mathrm{e}^{-\mathrm{i}\omega_k t}\,\mathrm{e}^{\mathrm{i}\vec{k}_\perp\cdot\vec{r}}(\,\mathrm{e}^{\mathrm{i}k_x x}-R_{ks}\,\mathrm{e}^{-\mathrm{i}k_x x})+\mathrm{c.c.}\Big]$$
$$+\frac{1}{(2\pi)^3}\int_{-\infty}^{\infty}\mathrm{d}^2\vec{k}_\perp\Big[\int\limits_{\mathrm{i}k_\perp\to 0\to\infty}\mathrm{d}k_x\Big(\frac{\hbar}{2\omega_k\varepsilon_0}\Big)^{1/2}\vec{e}_s b_{kp}\mathrm{i}k\,\mathrm{e}^{-\mathrm{i}\omega_k t}\,\mathrm{e}^{\mathrm{i}\vec{k}_\perp\cdot\vec{r}}(-\,\mathrm{e}^{\mathrm{i}k_x x}+R_{kp}\,\mathrm{e}^{-\mathrm{i}k_x x})+\mathrm{c.c.}\Big]$$
$$+\frac{1}{(2\pi)^3}\int_{-\infty}^{\infty}\mathrm{d}^2\vec{k}_\perp\Big[\int\limits_{-\infty\to 0\to -k_\perp}\mathrm{d}k_x\Big(\frac{\hbar}{2\omega_k\varepsilon_0}\Big)^{1/2}\vec{e}_p b_{ks}\mathrm{i}k\,\mathrm{e}^{-\mathrm{i}\omega_k t}\,\mathrm{e}^{\mathrm{i}\vec{k}_\perp\cdot\vec{r}}T^*_{ks}\,\mathrm{e}^{-\mathrm{i}k_x x}+\mathrm{c.c.}\Big]$$
$$+\frac{1}{(2\pi)^3}\int_{-\infty}^{\infty}\mathrm{d}^2\vec{k}_\perp\Big[\int\limits_{-\infty\to 0\to -k_\perp}\mathrm{d}k_x\Big(\frac{\hbar}{2\omega_k\varepsilon_0}\Big)^{1/2}\vec{e}_s b_{kp}\mathrm{i}k\,\mathrm{e}^{-\mathrm{i}\omega_k t}\,\mathrm{e}^{\mathrm{i}\vec{k}_\perp\cdot\vec{r}}T^*_{kp}\,\mathrm{e}^{-\mathrm{i}k_x x}+\mathrm{c.c.}\Big]\,, \tag{20}$$

where vectors $\vec{e}_s$ and $\vec{e}_p$ point, respectively, to the directions as $\vec{E}$ shown in Figs. 2 and 3. The integration of the k_x space can be changed to the integration of the frequency $\int\limits_{\mathrm{i}k_\perp\to 0\to\infty}k_x\,\mathrm{d}k_x=\int\limits_0^\infty(\omega/c^2)\,\mathrm{d}\omega$ with $\omega^2/c^2=k_x^2+k_\perp^2$, which coincides with Fourier time transform in Eq. (1).

The physical vacuum is given by $b_k|0\rangle=0$ and $\langle 0|b_k b_k^\dagger|0\rangle=1$. At finite temperature, the physical vacuum $|0\rangle$ should be replaced by the thermodynamical vacuum state $|\Omega\rangle$ with $\langle\Omega|b_k b_k^\dagger|\Omega\rangle=\coth(\hbar\omega_k/2k_BT)$, which represents a thermal Planck distribution including the zero-point term. From Eq. (20) and taking into account $R_{k\lambda}R^*_{k\lambda}+T_{k\lambda}T^*_{k\lambda}=1$, the mean expectation value of operator T_{xx} in region I will be

$$\langle\Omega\,|T_{xx}|\;\Omega\rangle_I=\frac{1}{2}\sum_\lambda\frac{\hbar c}{(2\pi)^3}\int\mathrm{d}k_y\int\mathrm{d}k_z\int\limits_{\mathrm{i}k_\perp\to 0\to\infty}\mathrm{d}k_x\frac{k_x^2}{\sqrt{k_x^2+k_y^2+k_z^2}}\coth\Big(\frac{\hbar\omega_k}{2k_BT}\Big)+\mathrm{c.c.} \tag{21}$$

Similarly for the region between the plates,

$$\langle\Omega\,|T_{xx}|\;\Omega\rangle_{III}=\frac{1}{2}\sum_\lambda\frac{\hbar c}{(2\pi)^3}\int\mathrm{d}k_y\int\mathrm{d}k_z\int\limits_{\mathrm{i}k_\perp\to 0\to\infty}\mathrm{d}k_x\frac{k_x^2}{\sqrt{k_x^2+k_y^2+k_z^2}}(C_{k\lambda}C_{-k\lambda}+D_{k\lambda}D_{-k\lambda})$$
$$\times\coth\Big(\frac{\hbar\omega_k}{2k_BT}\Big)+\mathrm{c.c.} \tag{22}$$

Finally we get the expression for the interaction between two parallel plates of unit area,

$$f=\frac{1}{2}\mathrm{Re}\sum_\lambda\frac{\hbar c}{(2\pi)^3}\int\mathrm{d}k_y\int\mathrm{d}k_z\int\limits_{\mathrm{i}k_\perp\to 0\to\infty}\mathrm{d}k_x\frac{k_x^2}{\sqrt{k_x^2+k_y^2+k_z^2}}(1-C_{k\lambda}C_{-k\lambda}-D_{k\lambda}D_{-k\lambda})\coth\Big(\frac{\hbar\omega_k}{2k_BT}\Big)\,. \tag{23}$$

Let us firstly discuss the Casimir effect between two perfect metallic plates, which means the reflection coefficients $|r_{d\lambda}|\to 1$ and $|r_\lambda|\to 1$. Setting $2k_x a=\mu$, the term in the brackets of the above formula is nonzero only for $\mu_n=2\pi n$, $n=0,1,2,\ldots$ For μ near resonances we may write

$$\lim_{r_\lambda\to 1}(C_{k\lambda}C_{-k\lambda}+D_{k\lambda}D_{-k\lambda})=\lim_{r_\lambda\to 1}\frac{1-r_\lambda^4}{1+r_\lambda^4-2r_\lambda^2\cos\mu}\approx 2\pi\delta(\mu-\mu_n)\,. \tag{24}$$

In this limit the Casimir force is

$$F_c=\frac{\hbar c}{\pi^3}\int_0^\infty\mathrm{d}k_x\int_0^\infty\mathrm{d}k_y\int_0^\infty\mathrm{d}k_z\frac{k_x^2}{\sqrt{k_x^2+k_y^2+k_z^2}}[1-2\pi\delta(\mu-\mu_n)]\coth\Big(\frac{\hbar\omega_k}{2k_BT}\Big)=\frac{k_BT}{4\pi a^3}{\sum_{l=0}^\infty}'\int\limits_{2a\xi_l/c}^\infty\frac{y^2\,\mathrm{d}y}{\mathrm{e}^y-1}\,. \tag{25}$$

Here the prime adds a multiple 1/2 near the term $l=0$ and $\xi_l=2\pi k_BTl/\hbar$ is the Matsubara frequencies. Since $k_x=n\pi/a$ takes discrete value only, the evanescent modes will not contribute to the Casimir effect in the case of the perfect metallic plates. However the evanescent modes makes an important role in the case of non-perfectly-reflecting metal. Using the relation $r_{k\lambda}r_{-k\lambda}+t_{k\lambda}t_{-k\lambda}=1$, equation (23) leads to

$$f=\mathrm{Re}\sum_\lambda\frac{\hbar c}{2\pi^2}\int_0^\infty\mathrm{d}k\int\limits_{1\to 0\to\mathrm{i}\infty}\mathrm{d}p\cdot k^3p^2\frac{1}{1-r_\lambda^{-2}(k)\,\mathrm{e}^{-2\mathrm{i}kpa}}\coth\Big(\frac{\hbar\omega_k}{2k_BT}\Big)\,. \tag{26}$$

The integration of p over $(0,1)$ represents the contribution of the oscillatory modes while the integration over the imaginary axis represents the contribution of the evanescent modes, which was recognized previously.[23,24]

In order to compare with the Lifshitz result, we change the integral path in the k and p complex plane and use the residue theorem as done by Lifshitz.[6,25,26] Noting the relations $r_{ds}=(K-\varepsilon)/(K+\varepsilon)$ and $r_{dp}=(K-p\varepsilon)(K+p\varepsilon)$

with $K=\sqrt{\varepsilon(\mathrm{i}\,\xi)+p^2-1}$, it is easy to obtain the Casimir force per unit area between two plates,

$$f=\frac{k_B T}{\pi c^3}{\sum_{l=0}^{\infty}}'\xi_l^3\int_1^{\infty}p^2\,\mathrm{d}p\Big[\frac{1-Q_1(\mathrm{i}\,\xi_l)}{Q_1(\mathrm{i}\,\xi_l)}+\frac{1-Q_2(\mathrm{i}\,\xi_l)}{Q_2(\mathrm{i}\,\xi_l)}\Big], \tag{27}$$

where a more detailed representation for the function $Q_{1,2}$ is[1,13]

$$Q_1(i\xi)=1-\Big[\frac{(K-\varepsilon p)(K+\varepsilon p)-(K-\varepsilon p)(K+\varepsilon p)\,\mathrm{e}^{-2\xi Kd/c}}{(K+\varepsilon p)(K+\varepsilon p)-(K-\varepsilon p)(K-\varepsilon p)\,\mathrm{e}^{-2\xi Kd/c}}\Big]\mathrm{e}^{-2\xi pa/c},$$

$$Q_2(i\xi)=1-\Big[\frac{(K-p)(K+p)-(K-p)(K+p)\,\mathrm{e}^{-2\xi Kd/c}}{(K+p)(K+p)-(K-p)(K-p)\,\mathrm{e}^{-2\xi Kd/c}}\Big]\mathrm{e}^{-2\xi pa/c}. \tag{28}$$

For $T\to 0$ in the sense that $k_BT\ll\hbar c/a$ the sum over l turns to an integral

$$f_{T=0}=\frac{\hbar}{2\pi^2c^3}\int_1^{\infty}p^2\,\mathrm{d}p\int_0^{\infty}\xi^3\,\mathrm{d}\,\xi\Big[\frac{1-Q_1(\mathrm{i}\,\xi_l)}{Q_1(\mathrm{i}\,\xi_l)}+\frac{1-Q_2(\mathrm{i}\,\xi_l)}{Q_2(\mathrm{i}\,\xi_l)}\Big]. \tag{29}$$

The above expression coincides with Lifshitz result[1,6] for the force per unit area between semispaces with a dielectric permittivity ε. To obtain this limiting case from Eq. (27) one should put $d\to\infty$,

$$f_{\mathrm{Lifshitz}}=\frac{k_BT}{\pi c^3}{\sum_{l=0}^{\infty}}'\xi_l^3\int_1^{\infty}p^2\,\mathrm{d}p\Big\{\Big[\Big(\frac{K(\mathrm{i}\,\xi_l)+p}{K(\mathrm{i}\,\xi_l)-p}\Big)^2\mathrm{e}^{2\xi_l pa/c}-1\Big]^{-1}+\Big[\Big(\frac{K(\mathrm{i}\,\xi_l)+p\varepsilon}{K(\mathrm{i}\,\xi_l)-p\varepsilon}\Big)^2\mathrm{e}^{2\xi_l pa/c}-1\Big]^{-1}\Big\}. \tag{30}$$

It can also be obtained directly from Eq. (26) by replacing r_λ with its mean value $r_{d\lambda}$ due to its rapid vibration since $d\gg a$.

4 Summary and Discussion

We have generalized the Kupisewska's method and presented another derivation of the Lifshitz results starting directly from the quantization of the electromagnetic field in the presence of dielectrics, where the physical meaning of 'evanescent mode' is discussed. The Casimir force between two dielectric plates can be regarded as the pressure of the electromagnetic vacuum, which characterized by the quantum vacuum mode functions. The calculation of the Casimir force needs to use all the vacuum mode functions, which include the contribution of the evanescent modes. Only in the case of the perfect metallic plates, will the evanescent modes become unimportant, since the metallic plates discrete the modes $k_x=n\pi/a$.

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